

Introduction to Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI architecture that combines information retrieval with large language models. Instead of depending only on pre-trained knowledge, RAG systems retrieve relevant information from external data sources such as PDFs, websites, databases, and APIs. The RAG pipeline generally consists of several stages including data ingestion, preprocessing, chunking, embedding generation, vector database storage, retrieval, and response generation. Data ingestion is the process of collecting and loading data into the system. The data can be structured, unstructured, or semi-structured. Examples include CSV files, PDFs, emails, HTML pages, and API responses. Preprocessing improves data quality by removing noise, handling formatting issues, and cleaning unwanted content. After preprocessing, documents are divided into smaller chunks so that embedding models and LLMs can process them efficiently. Embeddings are numerical vector representations of text that capture semantic meaning. These embeddings are stored in a vector database such as FAISS or ChromaDB. During retrieval, similarity search is performed between the user query and stored embeddings. RAG systems help improve response accuracy, reduce hallucinations, and provide access to up-to-date information.

Common challenges in RAG systems include noisy data, duplicate documents, OCR extraction issues, large document handling, and multilingual datasets. Proper chunking and preprocessing play a major role in improving retrieval quality. Popular tools used in RAG pipelines include LangChain, PyPDF, Beautiful Soup, FAISS, ChromaDB, and OpenAI embedding models.